Empirical Proof Record: VALIDATED

2026-05-21T23:23:41.453982+00:00

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Proof ID: `732f9c207069c1d6a75cd1bba5603414…`
Created: 2026-05-21T23:23:41.453982+00:00

1. Claim

> Across a trajectory with spaced re-exposures, the substrate keeps recognising every re-exposed stimulus: (for every same-stimulus re-exposure pair (i, j) in the trajectory, Jaccard(concepts_i, concepts_j) $\geq$ 0.80). The reported floor is the worst pair across the whole run.

- **Operationalisation:** Run a schedule of len(stimuli) unique stimuli each repeated 3× and interleaved (gap = len(stimuli)) through Kimera's entity target (Takwin). For each stimulus, take the per-cycle "concepts" set at each exposure and compute Jaccard over all exposure pairs. recognition_jaccard_floor = min over all pairs across all stimuli; the LTL invariant holds iff floor $\geq \theta$.
- **Threshold:** `recognition_jaccard_floor >= 0.8 jaccard`
- **H0:** H0: floor < 0.80 — recognition collapses under manifold deformation somewhere in the run
- **H1:** H1: floor $\geq$ 0.80 — recognition is stable across the whole trajectory (memory-as-deformation holds as a flow property)

2. Verdict

VALIDATED — observed `0.8695652173913043`
recognition floor 0.8696 over 18 re-exposure pairs (8 stimuli × 3 exposures); mean Jaccard 0.9754; 6 exposures produced no concept set; worst pair: stimulus 0 cycles 0↔8 = 0.8696

3. Evidence

Pillar	Statistic	Value	95% CI	Lib
`O.flow.recognition_stability`	`recognition_jaccard_floor`	0.8695652173913043	—	oph

4. Pre-registration

- Registered at: `2026-05-21T23:23:41.071327+00:00`
- Config hash: `0fbe8ce56b2a2db7203910b72aaebc0c…`
- Data hash: `3c5602a31f04f6ccaa77bd90fa1416a6…`

5. Signature

`e8454a9c875d1d1cdc9c3e9821d1b9556de3d2bbad9241f9206a1ec58282a563`